

# On the Trade-offs of LLM-Augmented Multimodal Foundation Models for Clinical Image Classification

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## Motivation

- Clinical Challenge
  - MedGemma can generate pseudo-reports from images
  - Multi-label classification remains difficult under class imbalance
- Opportunity of Multimodal LLM Augmentation
  - BiomedCLIP underuses its text branch at inference
  - MedGemma pseudo-reports may enhance image-based prediction
- Need for Reliable Evaluation
  - Generated text may hallucinate or miss findings
  - Compare image-only, LLM-augmented, and LLM-only settings using both accuracy and safety metrics

## Pseudo-Report Multimodal Fusion

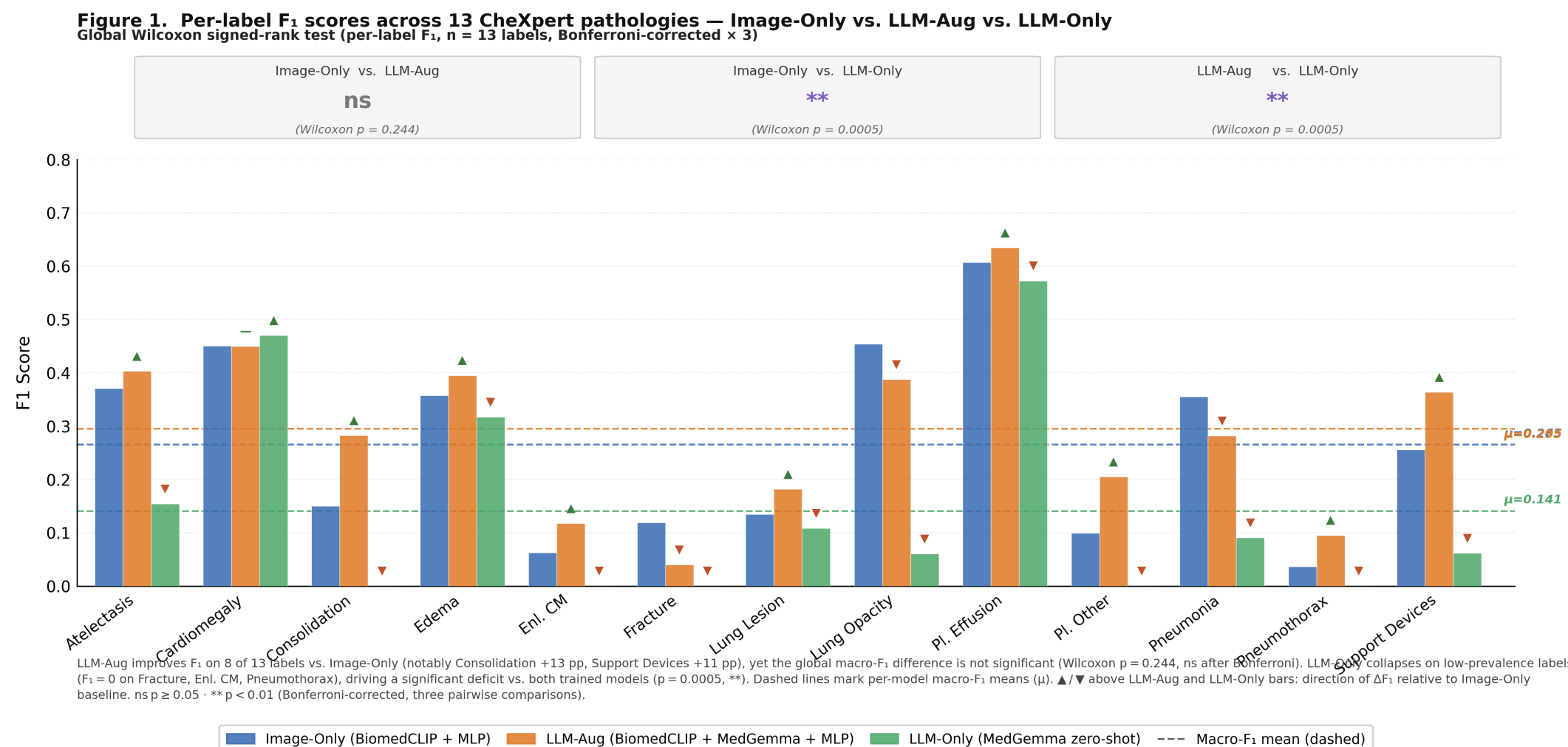
- Three-Way Evaluation Framework.
  - Image-only BiomedCLIP
  - LLM-augmented multimodal model
  - LLM-only zero-shot baseline
- Pseudo-Report Generation
  - Use MedGemma to generate chest X-ray pseudo-reports from input images
  - No ground-truth reports used at inference
- Multimodal Feature Fusion
  - Extract image embeddings and text embeddings with BiomedCLIP
  - Concatenate fused features for 13-label classification with an MLP head
- Two-Phase Training
  - Phase 1: Train classifier with frozen backbone
  - Phase 2: End-to-end fine-tuning with lower learning rate
- Reliability-Aware Evaluation
  - Standard metrics: F1, ROC-AUC, sensitivity, specificity
  - Safety metrics: Fabrication Error Rate (FER) and Omission Rate (OR)

## Model Optimization

- Weighted Multi-Label Learning
  - Optimize 13 pathology labels using sigmoid + binary cross-entropy loss
  - Apply weighted sampling to address imbalance
- Optimization Goal
  - Learn robust fused image-text features
  - Improve overall classification reliability and sensitivity-specificity balance

## Per-Label Analysis

- Overall Trend
  - LLM augmentation improved mean per-label F1 over the image-only baseline
  - LLM-only performance dropped markedly on most labels
- Labels with Strong Gains
  - Better performance on visually explicit findings and consistently described abnormalities
  - Notable improvements for three labels:
    - Consolidation
    - Pleural findings
    - Support devices
- Clinical Interpretation
  - Generated text helps common, obvious findings
  - Reliable handling of subtle pathologies remains a key challenge.



## Contributions

- Introduced an LLM-Augmented CLIP Framework: Combined BiomedCLIP image embeddings with MedGemma pseudo-reports for chest X-ray classification.
- Conducted a Systematic Three-Way Benchmark: Compared image-only, LLM-augmented, and LLM-only settings on 13-label chest X-ray prediction
- Revealed a Key Trade-off: LLM augmentation improved specificity and agreement metrics, but reduced sensitivity due to missed findings
- Proposed Reliability-Aware Evaluation: Introduced Fabrication Error Rate (FER) and Omission Rate (OR) to measure hallucination impact on downstream classification

## Key Performance

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Algorithm: Pseudo-Report Generation as Augmentation for LLM Classification
Require: CXR image I
1: Train BiomedCLIP-ALR grounder G (bounding box + label predictor);
2: MedGemma model M;
3: BiomedCLIP image encoder f_i: R^{H \times W \times 3} \rightarrow R^{512};
4: BiomedCLIP text encoder f_t: text \rightarrow R^{512};
5: Train MLP classifier g: R^{1024} \rightarrow R^{13};
6: Pathology label set C = {c_1, ..., c_{13}};
7: Decision threshold \tau = 0.5
Ensure: Per-label binary predictions y \in \{0, 1\}^{13};
8: Structured pseudo-report R;
9: Verified clinical summary R' (Post 2)

10:
11: z_i \leftarrow f_i(I) \quad \triangleright \text{Phase 1 --- Visual Grounding (BiomedCLIP-ALR)}
12: \{(b_i, l_i, h_i)\}_{i=1}^N \leftarrow G(I) \quad \triangleright \text{Predict bounding boxes, anatomical labels, confidence scores}
13: Overlay bounding boxes (b_i) onto I to produce annotated image I' \quad \triangleright \text{Phase 2 --- Pseudo-Report Generation (MedGemma, Pass 1)}
14: Construct structured prompt P: \quad P \leftarrow [f', \{(l_i, b_i, h_i)\}_{i=1}^N]
\quad \triangleright \text{Instructing M to describe severity, laterality, and clinical significance}
15: for each detected region:
16: \quad R \leftarrow M(P) \quad \triangleright \text{MedGemma generates free-text pseudo-report conditioned on grounded findings}
17: Parse R into structured JSON J_R;
\quad J_R = \{ \text{pathology}_i, \text{anatomy}_i, b_i, \text{severity}_i, \text{laterality}_i, \text{descriptor}_{i,j} \}_{i=1}^N

18:
19:
20: z_t \leftarrow f_t(R) \quad \triangleright \text{Phase 3 --- Embedding Fusion and Classification}
21: z_i \leftarrow z_i / \|z_i\|_2; \quad z_t \leftarrow z_t / \|z_t\|_2 \quad \triangleright \text{Encode full pseudo-report text via BiomedCLIP text encoder}
22: z_{fused} \leftarrow [z_i, z_t] \in R^{1024} \quad \triangleright z_i, z_t \geq 12 \text{ normalize both embeddings}
23: y \leftarrow g(z_{fused}) \in R^{13} \quad \triangleright \text{MLP classifier with sigmoid activation}
24: y \leftarrow 1[y \geq \tau] \quad \triangleright \text{Apply threshold } \tau = 0.5 \text{ for binary multi-label prediction}

25:
26: Send J_R and y to reasoning LLM L (e.g. GPT-4.5 or Claude Sonnet 4.5)
27: for each finding pair (c_i, c_j) \in y with b_i = b_j = 1 do
28: \quad Check clinical co-occurrence plausibility of (c_i, c_j)
29: \quad If co-occurrence contradicted by L then
30: \quad \quad Flag inconsistency, adjust lower-confidence label
31: \quad end if
32: end for
33: R' \leftarrow L(J_R, y, "generate verified radiology report") \quad \triangleright \text{Synthesize final clinically consistent narrative}
34: Compute hallucination-aware metrics:
\quad FER = \frac{FP}{FP + TP}; \quad OR = \frac{FN}{FN + TP}

35: return y, R, R', FER, OR
```

| Metric              | Image-Only   | LLM-Aug                    | LLM-Only                   |
|---------------------|--------------|----------------------------|----------------------------|
| Macro F1            | 0.265        | <b>0.295<sup>ns</sup></b>  | 0.141 <sup>**</sup>        |
| Micro F1            | 0.295        | <b>0.388<sup>***</sup></b> | 0.288 <sup>***</sup>       |
| Hamming Acc.        | 0.620        | 0.818 <sup>***</sup>       | <b>0.892<sup>***</sup></b> |
| Exact Match Acc.    | 0.156        | 0.257 <sup>***</sup>       | <b>0.284<sup>***</sup></b> |
| Sensitivity (Macro) | <b>0.735</b> | 0.456 <sup>***</sup>       | 0.127 <sup>***</sup>       |
| Specificity (Macro) | 0.603        | 0.839 <sup>***</sup>       | <b>0.967<sup>***</sup></b> |
| Youden's J          | <b>0.338</b> | 0.295 <sup>ns</sup>        | 0.094 <sup>**</sup>        |
| ROC-AUC (Macro)     | <b>0.738</b> | 0.729 <sup>ns</sup>        | —                          |
| ROC-AUC (Micro)     | 0.755        | <b>0.783<sup>ns</sup></b>  | —                          |
| FER                 | 0.817        | 0.704 <sup>***</sup>       | <b>0.557<sup>***</sup></b> |
| FER <sub>m</sub>    | 0.773        | 0.647 <sup>***</sup>       | <b>0.441<sup>***</sup></b> |
| OR                  | <b>0.225</b> | 0.437 <sup>***</sup>       | 0.786 <sup>***</sup>       |

- LLM-Augmented Best Overall Balance
  - Highest Micro-F1 (0.388), Hamming Accuracy (0.818), and Exact-Match Accuracy (0.257)
- Image-Only Best Sensitivity
  - Highest Macro Sensitivity (0.735), detecting more true abnormalities
  - But showed lower specificity and more false positives
- LLM-Augmented Improves Precision
  - Specificity increased from 0.603 to 0.839.
  - Fabrication Error Rate (FER) reduced from 0.817 to 0.704.
- Trade-off: More Missed Findings
  - Omission Rate increased from 0.225 to 0.437 under LLM augmentation
  - Indicates fewer false alarms but more missed positives
- LLM-Only Most Conservative
  - Highest specificity (0.967) and lowest FER (0.557)
  - However, very low sensitivity (0.127) and weakest Macro-F1 (0.141)

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